

NOTE: This worksheet is divided into two parts (Part I and Part II) and it is intended that the first part is done following the slides of the theoretical and practical lectures. In the second part it is intended that the student perform the exercises implementing the subjects addressed and practiced in the first Part. The student must refer to the slides provided to carry out this worksheet.

Part I

Exercise 1

implement a `LinkedBinarySearchTree` as suggested in the lecture slides (note that `removeAllOccurrences` and `removeMin` methods are not yet implemented). Demonstrate the use of `LinkedBinarySearchTree` through a setting of your choice.

Notes:

- The `removeAllOccurrences` method should remove all occurrences of a particular element passed by parameter.
- The `removeMin` method should remove the lowest tree element.

Exercise 2

Implement a `ArrayBinarySearchTree` (which is like the previous exercise but uses an array instead of a linked list). Demonstrate the use of `ArrayBinarySearchTree` through a setting of your choice (the same defined in exercise 1).

Part II

Exercise 1

What is a binary search tree?

Exercise 2

What do you understand by a degenerated binary tree?

Exercise 3

Implement an ordered list using a binary search tree.

Note: Please note that you must meet certain requirements.

Exercise 4

Implement an AVL tree (AVL Tree).

Note: Attention to the balancing factor: *"The AVL trees keep up with the difference in height between the right and left sub-trees of each node."*